

Kishal Tandel

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Undergraduate Physics Student | Computational Modeling of Quantum Systems

Education

- August, 2023 – present **BS-MS in Physical Sciences**, Indian Institute of Science Education and Research, Kolkata, India
CGPA : 8.14/10 (after 5th Semester)
- 2023 **ISC**, METAS M.C.D. School of S.D.A. , Surat, Gujarat, India
Class 12 - 93.25%, Indian School Certificate (ISC)
- 2021 **ICSE**, METAS M.C.D. School of S.D.A. , Surat, Gujarat, India
Class 10 - 95.40%, Indian Certificate of Secondary Examination (ICSE)

Research Interests

Computational quantum physics, electronic transport in low-dimensional systems, and numerical modeling of dynamical physical systems.

I am particularly interested in Hamiltonian-based modeling and simulation-driven exploration of emergent behavior in quantum and nonlinear systems.

Research Work

- April 14, 2025 **One-Dimensional Quantum Tunneling**
Derived analytical solutions of the 1D time-independent Schrödinger equation across all energy regimes and computed transmission and reflection coefficients. Numerically visualized continuity at $E = V_0$ using Python and compared quantum and classical transport behavior.

Projects

- May, 2025 **Chaotic Trio**
Built a fourth-order Runge–Kutta simulation framework in C++ to model planar three-body gravitational dynamics and analyze sensitivity to initial conditions. Generated visualizations demonstrating onset of chaotic orbital behavior.

Relevant Skills

Numerical Modeling of Physical Systems

Differential Equation Solvers (RK4)

Quantum System Simulation

Scientific Visualization using Python

Skills

Programming Python (Scientific Computing), C++, C, Java

Language English, Hindi, Gujarati

Homepage kishaltandel.github.io